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Bid Shading in programmatic advertising: all you need to know

Explore the depths of bid shading with our latest blog article. Gain insights into its role in programmatic advertising, deciphering the shift of second-price to first-price auctions, and understanding the algorithms driving bid shading strategies. Uncover the benefits and drawbacks of bid shading and its implications for publishers.



Bid shading, a technique employed by demand-side platforms (DSPs) and some SSPs in ad tech, has evolved significantly over the years. Initially, it was a concern for publishers due to its potential to reduce earnings, but lately the landscape has changed.

With advancements in auction strategies and the adoption of dynamic flooring, publishers now have effective tools at their disposal to **mitigate the financial impacts of bid shading**.

Artificial Intelligence and machine learning, have allowed publishers to maintain more control over their floor pricing, ensuring that bid shading no longer poses the substantial challenge it once did.

In this article, we'll delve into what is bid shading, how bid shading works, the bid shading algorithm, and how it impacts publishers' ability to generate revenue from their advertising inventory.

What is bid shading?

Bid shading is a technique used by demand-side platforms (DSPs) and some SSPs in ad tech, to **adjust winning bids in favor of media buyers**.

It became an industry practice after the trend towards first-price auctions. However, when bid shading was less common, buyers were uncertain whether they were participating in first or second-

Second-price auction's ambiguity initially benefited publishers with higher earnings, but advertisers were dissatisfied with the bidding rates in the emerging first-price auction environment.

That's why DSPs came up with bid shading to estimate the bid that would have succeeded in a second-price auction. This allows them to place competitive bids without reducing their chances of winning.

While this strategy can lead to up to 20% savings for buyers, it often results in [reduced earnings](#) for publishers.

While bid shading is applied in various programmatic advertising auctions, it's worth noting that not every demand-side platform incorporates this technique. Instead, bid shading stands as one additional tool available in the publisher's [ad tech stacks](#).

How does bid shading work?

Bid shading is a predictive algorithm used by demand-side platforms (DSPs) and SSPs, to **define the best bid price for an impression in first-price auctions**. This means that it curbs price surges when shifting from second-price to first-price auctions. In the long run, it benefits both publishers and advertisers - even though in the short term, it often results in reduced CPMs for publishers.

So, when bid-shading intermediaries **move from a second-price to a first-price auction**, the highest bidder for premium inventory pays just a bit more (one cent) than the second-highest bid. As it directly affects the return on ad spend, buyers generally use bid shading to pay less for impressions.

With the adoption of bid shading, many publishers find their Cost Per Mille (CPM) reverting to what it would be in a second-price auction.

This approach includes tools like [Google's DV 360](#), [The Trade Desk](#), and others. When setting the bid, bid shading algorithms consider factors like pricing data, site, ad size, exchange, competitive dynamics, etc., to set the winning bid so that media buyers don't pay too much. The ideal bid price is high enough to beat other bids but ranges between the first and second-highest bidders.

Bid shading auction: first price auction vs. second price auction

Understanding bid shading requires a grasp of the different auction models common in programmatic advertising – the first price auction and the second price auction.

First-price auction

Notably, **impressions are valued based on what buyers are willing to pay**, causing advertisers in first-price auctions to worry about potential overpayment. In these auctions, **if an advertiser bids \$20 and it's the highest, they pay \$20**, no matter the other bids.

In response, publishers in first-price auctions can set a minimum bid (a **floor price**) to guarantee a certain amount from the auction. That ensures their **inventory for real-time bidding (RTB) doesn't go below that floor price**. This assures publishers that winning bids meet the minimum threshold in their supply-side platforms (SSPs).

To combat further decreases in CPMs due to bid shading and protect their inventory, publishers normally deploy specific [dynamic flooring tools](#) built for first and second-price auctions.

Just like buyers, they analyze their auction's bid history to set informed starting bid suggestions.

These tools support major DSP and SSP integrations, providing transparency and better results to demand-side platforms (DSPs), something not achievable with [Google Ad Manager](#) (GAM) and [unified pricing rules](#) (UPRs).

Second-price auctions

Programmatic advertising frequently employs **second-price auctions**, primarily because they empower advertisers to **customize their bidding strategies** for optimal use of their marketing budget.

In these auctions, advertisers can submit bids of any amount, but if they emerge as the winners, they only need to pay \$0.01 more than the second-highest bidder. Let's see a practical example.

- **First price auction:** let's say DSP A bids with a price of \$10 and DSP B bids with a price of \$8. If this was a first-price auction, the DSP A would win, and they would pay \$10.
- **Second price auction:** in a second price auction, DSP A would also win, but they would pay only one cent higher than the second highest bid which is the dollars, so they would pay \$8.01.

Notably for publishers, the floor pricing strategy proves more advantageous in second-price auctions, as it plays a role in narrowing the difference between the highest bid and the clearing price.

Bid shading explained: what are bid shading methods?

Bid shading relies on a predictive algorithm that **analyzes historical data and real-time auction dynamics**. Advertisers use this algorithm to fine-tune their bids, striking a balance between winning

When DSPs implement bid shading, they **analyze past winning bids to calculate an average** between the lowest and highest bids on a given impression. This aims to prevent advertisers from overpaying for ad space.

The bid shading process varies among ad tech vendors, aiming to find a bid price between the first-price and second-highest bid. Two methods have been employed in the last few years:

Median method

This bid shading method is more outdated and takes the median between the highest and second-highest bid to split the difference between the first price and the second price auction and calculate a discount on a given bid.

Historical data method: the modern approach

The second method consists of examining **historical yield data**, including **win rates, ad unit size, bidding history, and high click-through rates**. The auction platform is going to automatically analyze the historical price that the winning bidder paid for a particular site slot across different ad dimensions, and then apply a discount to the final price.

Nils Lind, [Assertive Yield's CEO](#), explains: "Now, all of the bid optimization is based on historical data, which is good in one end because it offers a more precise evaluation of the market and the economics around it. The way a publisher can still squeeze out more revenue is really to focus their attention on qualified users."

While bid shading is offered as an add-on by some ad tech companies, others incorporate it into their services for a fee. Buyer pricing tools, such as [The Trade Desk's Koa](#), use bid shading to reduce costs in programmatic first-price auctions.

The Trade Desk claims a [20% reduction in publisher CPMs](#) with its tool, and Google also employs bid shading for its buyers. However, publishers are feeling that 20% average price reduction.

Several DSPs and exchanges, including major players like Google's DV360, offer optimized [fixed CPM bidding](#) as a free bid shading tool. At the same time, The Trade Desk charges a fee for Koa, pocketing a percentage of buyers' savings on each impression.

Does bid shading actually happen?

The prevalence of bid shading is a topic of discussion within the programmatic advertising sphere. Advertisers often question its effectiveness and whether it's a widely adopted practice. While bid shading exists, its frequency can vary based on factors such as industry, ad format, and campaign objectives.

"Deciding the proper amount to bid on the auction depends on knowledge of the exchange, sites, and many other variables that may also change over time. Getting it right can save a lot of money while getting it wrong can mean you lose opportunities to bid on valuable inventory," said Ari Paparo, CEO of Beeswax, [to the AdExchanger](#).

Some industry experts also deem bid shading a temporary solution because DSPs have built their own first-price algorithms.

Bid Shading benefits and drawbacks

Benefits

Bid shading particularly benefits advertisers by giving them **more control over ad space and their budgets**.

Its algorithms also assess the impact of bid pricing on the win rate, helping bidders optimize their bids for CPM, CPC, CPA, and other metrics. Bid shading also **saves time by automating the calculation of lower costs**. This upfront information in the auction model assures advertisers that they are not overpaying for ad inventory.

For **publishers**, although they are going to get a **slightly lower price and lower revenue in what they would earn**, even these averaged bids tend to be higher than second-price auction bids.

Also, because the publisher is providing something of a discount to DSPs, it is more likely they will have more DSPs competing in the auction for their inventory. Ultimately, that can boost their revenue in the long run.

At the same time, publishers can maintain price control and manage yields by setting [smart price floors in real-time](#). This can be done on the impression level with the user's CPM information and **dynamically adjusting to market value**.

Drawbacks

While bid shading comes with advantages, it also has drawbacks, it tends to favor buyers over publishers. The **analysis of historical data helps prevent advertisers from placing high bids** to secure inventory, protecting them from overpaying. However, **this reduction in bid amounts impacts publishers**, leading to lower earnings.

Another significant drawback for all parties involved in bid shading is the **lack of transparency**. Experts argue that bid shading takes a step backward in regards to the first price auction introducing a lack of transparency back into the auction mechanics, whereas the first price is trying to make it fully transparent

Buyers face challenges in assessing attribution and gauging how effectively their tech partner is implementing bid shading. This poses a hurdle for evaluating the efficiency of bid shading.

Strategies for publishers in the face of bid shading

At this point, it's clear that bid shading impacts both advertisers and publishers. As publishers adapted to first-price auctions, they also refined their partnerships.

"The only thing we (publishers) can do is to ensure that we have as much competition as possible where it matters, but also to basically reduce the amount of partners where it doesn't matter. It's about having very high quality standards with SSPs", says Nils Lind.

This approach emphasizes the importance of high standards in ad tech partnerships. Besides, another strategy publishers use is to deploy **dynamic flooring** to ensure they are reaching optimal revenue despite bid shading.

With the [right tools](#), publishers can implement smart floor prices to enable real-time market adjustments, safeguarding inventory against issues like low-quality ads, high operational expenses, and declining CPMs.

When publishers employ a good [bid request optimization tool](#), it automatically filters out: no-value bid requests, browsers, low-viewability, ad units, and sizes to increase their win rates with each SSP proving to be a valuable partner for them, enhancing their user experience while improving the SPO.

Avoid remaining bid shading impact with yield optimization

Advocates argue that bid shading establishes a fair price for ad units, even though it might fall below publishers' preferences. Therefore, publishers must grasp the dynamics of first-price auctions to

How can publishers leverage floor pricing and AI-powered optimization for continuous growth? The key is to protect inventory with intelligent bidding across Prebid, Amazon A9, Open Bidding, and AdX.

Protect your inventory and maximize your revenue, by analyzing bid history and market fluctuations in real-time with [a tool](#) that automatically sets dynamic floor pricing according to current market value, and empowers you to set intelligent starting bids, effectively countering bid shading's impact.

Using a control group to maintain unmodified floor prices or a publisher's existing strategy, it should also utilize machine-learning predictions to set smart floors on an impression level based on user CPM, taking into account factors like GEO, time of day, operating system, and user history.

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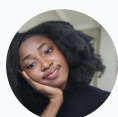
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